Chapter 8. Baire Spaces and Dimension Theory

The Baire category theorem, nowhere-differentiable functions, covering dimension and embeddings

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8.1 Baire spaces

Motivation. The Baire category theorem, a complete metric space is not a countable union of nowhere-dense sets, proves existence by showing the "bad" set is small.

$A \subset X$ has **empty interior** if it contains no nonempty open set. X is a **Baire space** if every countable union of closed sets with empty interior has empty interior.

Example 8.1

$\mathbb{Q}$ is not a Baire space: it is the countable union of its one-point sets, which are closed with empty interior. $\mathbb{Z}_+$ is a Baire space: every nonempty subset is open, so the only set with empty interior is $\emptyset$. The irrationals form a Baire space, since they admit a complete metric (they are homeomorphic to $\mathbb{Z}_+^\omega$; not proved here).

Lemma 8.1 (48.1)

X is a Baire space iff for every countable collection $\{U_n\}$ of dense open sets, $\bigcap U_n$ is dense.

Proof. U is open and dense iff $X - U$ is closed with empty interior, and $\bigcap U_n$ is dense iff $X - \bigcap U_n = \bigcup (X - U_n)$ has empty interior. ■

Theorem 8.2 (48.2, Baire category theorem)

If X is a compact Hausdorff space or a complete metric space, then X is a Baire space.

Proof. Let A_n be closed with empty interior and U_0 nonempty open; we find a point of U_0 outside every A_n. Since $U_0 \not\subset A_1$, pick $y \in U_0 - A_1$. X is regular (Theorem 4.8, or metric), so there is a neighbourhood U_1 of y with $\overline{U_1} \cap A_1 = \emptyset$ and $\overline{U_1} \subset U_0$; in the metric case choose also $\operatorname{diam} U_1 < 1$. Inductively, given nonempty open U_{n-1}, choose nonempty open U_n with $\overline{U_n} \subset U_{n-1}$, $\overline{U_n} \cap A_n = \emptyset$, and (metric case) $\operatorname{diam} U_n < 1/n$. The nested nonempty closed sets $\overline{U_n}$ have nonempty intersection: by the finite intersection property in the compact case, and by Lemma 8.3 in the complete case. A point of the intersection lies in U_0 and in no A_n. ■

Lemma 8.3 (48.3)

If $C_1 \supset C_2 \supset \cdots$ are nonempty closed sets in a complete metric space with $\operatorname{diam} C_n \to 0$, then $\bigcap C_n \neq \emptyset$.

Proof. Pick $x_n \in C_n$. For $n, m \geq N$, both lie in C_N, so $d(x_n, x_m) \leq \operatorname{diam} C_N$; the sequence is Cauchy and converges to some x. For each k, $x_n \in C_k$ for $n \geq k$, so $x \in C_k$. ■

Lemma 8.4 (48.4)

An open subspace Y of a Baire space X is a Baire space.

Proof. Let A_n be closed in Y with empty interior in Y, and $\overline{A_n}$ its closure in X, so $\overline{A_n} \cap Y = A_n$. If U is a nonempty open subset of X contained in $\overline{A_n}$, then U meets $A_n \subset Y$, and $U \cap Y$ is a nonempty open subset of Y contained in A_n, impossible. So $\overline{A_n}$ has empty interior in X. If U is open in Y (hence in X) and $U \subset \bigcup A_n \subset \bigcup \overline{A_n}$, then $U = \emptyset$ because X is Baire. ■

Theorem 8.5 (48.5)

Let X be a Baire space, (Y,d) metric, and $f_n : X \to Y$ continuous with $f_n(x) \to f(x)$ for every x. Then the set of points where f is continuous is dense in X.

Proof. For $N \in \mathbb{Z}_+$ and $\epsilon > 0$ let $A_N(\epsilon) = \{x : d(f_n(x), f_m(x)) \le \epsilon \ \forall n, m \ge N\}$, closed as an intersection of closed sets.

Step 1. every nonempty open U contains a nonempty open V such that, for some N, $d(f_n(x), f(x)) \le \epsilon$ for all $x \in V$, $n \ge N$. Indeed, since $(f_n(x))$ is Cauchy for each x, $X = \bigcup_N A_N(\epsilon)$; U is Baire (Lemma 8.4), so some $A_N(\epsilon) \cap U$ has nonempty interior V in U, and V is open in X. Letting $m \to \infty$ in the defining inequality gives the claim.

Step 2. let G_M be the union of all open V as in Step 1 for $\epsilon = 1/M$. It is open and, by Step 1, dense. So $C = \bigcap_M G_M$ is dense (Lemma 8.1). Let $x_0 \in C$ and $\epsilon > 0$; choose M with $1/M < \epsilon/3$, an open $V \ni x_0$ and N with $d(f_n(x), f(x)) \le 1/M$ on V for $n \ge N$, and a neighbourhood $W \subset V$ of x_0 with $d(f_N(x), f_N(x_0)) < \epsilon/3$ on W. Then for $x \in W$,

$$d(f(x), f(x_0)) \le d(f(x), f_N(x)) + d(f_N(x), f_N(x_0)) + d(f_N(x_0), f(x_0)) < \epsilon.$$

So f is continuous at each point of the dense set C. ∎

8.2 A nowhere-differentiable function

Motivation. By Baire's theorem most continuous functions are nowhere differentiable.

Theorem 8.6 (49.1)

Let $h : [0,1] \to \mathbb{R}$ be continuous and $\epsilon > 0$. There is a continuous, nowhere differentiable $g : [0,1] \to \mathbb{R}$ with $|h(x) - g(x)| < \epsilon$ for all x.

Proof. Let $I = [0,1]$ and $\mathcal{C} = \mathcal{C}(I, \mathbb{R})$ with the sup metric ρ; it is complete (Theorem 7.6), hence Baire. For $f \in \mathcal{C}$, $x \in I$ and $0 < k \le \frac{1}{2}$ let

$$\Delta f(x,k) = \max\Big\{\Big|\frac{f(x+k)-f(x)}{k}\Big|, \Big|\frac{f(x-k)-f(x)}{k}\Big|\Big\},$$

the maximum taken over those of $x \pm k$ lying in I (at least one does), and $\|\Delta_k f\| = \inf_x \Delta f(x,k)$. Let U_n be the set of f for which $\|\Delta_k f\| > n$ for some $k \le 1/n$.

(a) $\bigcap U_n$ consists of nowhere differentiable functions. If $f \in \bigcap U_n$, there are $k_n \le 1/n$ with $\Delta f(x, k_n) > n$ for all x. If f were differentiable at x (one-sidedly at an endpoint), the difference quotients would tend to $f'(x)$, so $\Delta f(x,k_n) \to |f'(x)|$, contradicting $\Delta f(x,k_n) > n$.

(b) U_n is open. Let $f \in U_n$ with $M = \|\Delta_k f\| > n$, and $\delta = (M-n)k/4$. If $\rho(f,g) < \delta$, then $|g(x \pm k) - g(x)| \ge |f(x \pm k) - f(x)| - 2\delta$, so $\Delta g(x,k) \ge \Delta f(x,k) - (M-n)/2$ and $\|\Delta_k g\| \ge (M+n)/2 > n$.

(c) U_n is dense. Let $f \in \mathcal{C}$ and $\epsilon' > 0$. By uniform continuity choose a partition of I on whose subintervals f varies by less than $\epsilon'/2$; the piecewise linear p interpolating f at the partition points satisfies $\rho(f,p) < \epsilon'/2$ ($p(x)$ is a convex combination of two values each within $\epsilon'/2$ of $f(x)$). Let L bound the slopes of p, so $|p(x) - p(y)| \le L|x-y|$. Choose m with $r = m\epsilon'/4 \ge L + n + 1$, and let ϕ be the sawtooth that is linear on each $[j/m, (j+1)/m]$, alternately rising from 0 to $\epsilon'/4$ and falling back, so $0 \le \phi \le \epsilon'/4$ and ϕ has slope $\pm r$. Put $g = p + \phi$, so $\rho(f,g) < \epsilon'$, and $k = \min(1/n, 1/2m)$. Each x lies in some $[j/m, (j+1)/m]$ of length $1/m \ge 2k$, so $x+k$ or $x-k$ lies in the same subinterval,

where $|\phi(x \pm k) - \phi(x)| = rk$. Then $|g(x \pm k) - g(x)| \geq rk - Lk$, so $\Delta g(x,k) \geq r - L \geq n+1$ for all x, and $g \in U_n$.

By Lemma 8.1, $\bigcap U_n$ is dense, so it contains some g with $\rho(h,g) < \epsilon$. ■

8.3 Introduction to dimension theory

Motivation. Covering dimension m means every open cover has a refinement of order $m+1$; every compact m-manifold embeds in $\mathbb{R}^{2m+1}$.

A collection $\mathcal{A}$ has **order** $m+1$ if some point lies in $m+1$ elements and no point lies in more than $m+1$. X is **finite dimensional** if for some m every open covering has an open refinement covering X of order at most $m+1$; the least such m is the **topological dimension** $\dim X$. It is a topological invariant. Every compact subspace of $\mathbb{R}$ has dimension at most 1; an arc has dimension 1.

Theorem 8.7 (50.1)

If X is finite dimensional and $Y \subset X$ is closed, then Y is finite dimensional and $\dim Y \leq \dim X$.

Proof. Let $m = \dim X$ and $\mathcal{A}$ a covering of Y by sets open in Y; choose A' open in X with $A' \cap Y = A$. Cover X by the A' and $X - Y$, take an open refinement $\mathcal{B}$ of order at most $m+1$ covering X, and intersect its elements with Y: this refines $\mathcal{A}$, covers Y, and has order at most $m+1$. ■

Theorem 8.8 (50.2)

If $X = Y \cup Z$ with Y, Z closed and finite dimensional, then $\dim X = \max\{\dim Y, \dim Z\}$.

Proof. Let $m = \max\{\dim Y, \dim Z\}$; $\dim X \geq m$ by Theorem 8.7.

Step 1. every open covering $\mathcal{A}$ of X has an open refinement covering X in which each point of Y lies in at most $m+1$ elements. Take a refinement $\mathcal{C}$ of $\{A \cap Y\}$ by sets open in Y, covering Y, of order at most $m+1$. For $C \in \mathcal{C}$ choose U_C open in X with $U_C \cap Y = C$ and $A_C \in \mathcal{A}$ with $C \subset A_C$. The collection $\{U_C \cap A_C\} \cup \{A - Y : A \in \mathcal{A}\}$ works: it covers X, refines $\mathcal{A}$, is open since Y is closed, and a point $y \in Y$ in $U_C \cap A_C$ lies in C.

Step 2. given $\mathcal{A}$, apply Step 1 to get $\mathcal{B}$ (good at points of Y), then Step 1 for Z to $\mathcal{B}$ to get $\mathcal{D}$ (good at points of Z). Choose $f : \mathcal{D} \to \mathcal{B}$ with $D \subset f(D)$ and let $E(B) = \bigcup\{D : f(D) = B\}$. The sets $E(B)$ are open, cover X and refine $\mathcal{A}$. If x lies in distinct $E(B_1), \ldots, E(B_k)$, then x lies in distinct $B_1, \ldots, B_k$ and in distinct $D_1, \ldots, D_k$ with $f(D_i) = B_i$; so $k \leq m+1$ whether $x \in Y$ or $x \in Z$. ■

Corollary 8.9 (50.3)

If $X = Y_1 \cup \cdots \cup Y_k$ with each Y_i closed and finite dimensional, then $\dim X = \max\{\dim Y_1, \ldots, \dim Y_k\}$.

Proof. Induction on k using Theorem 8.8. ■

Points $x_0, \ldots, x_k$ of $\mathbb{R}^N$ are **geometrically independent** if $\sum a_i x_i = 0$ and $\sum a_i = 0$ imply all $a_i = 0$; equivalently, no x_j lies in the plane P spanned by the others, $P = \{\sum t_i x_i : \sum t_i = 1\}$. A set of points of $\mathbb{R}^N$ is in **general position** if each subset of at most $N+1$ of its points is geometrically independent.

Lemma 8.10 (50.4)

Given $x_1, \dots, x_n \in \mathbb{R}^N$ and $\delta > 0$, there are $y_1, \dots, y_n$ in general position with $|x_i - y_i| < \delta$.

Proof. Induction: let $y_1 = x_1$, and suppose $y_1, \dots, y_p$ are in general position. Each subset T of at most N of them spans a plane $P(T)$ of dimension at most $N-1$, a closed set with empty interior. A finite union of closed sets with empty interior has empty interior (if $U \subset A \cup B$ is open and nonempty, then $U - A$ is a nonempty open subset of B). So choose y_{p+1} within δ of x_{p+1} outside every $P(T)$. A subset S of at most $N+1$ of $y_1, \dots, y_{p+1}$ is independent: either $S \subset \{y_1, \dots, y_p\}$, or $S = T \cup \{y_{p+1}\}$; then if $\sum_T a_i y_i + a y_{p+1} = 0$ with $\sum a_i + a = 0$ and $a \neq 0$, $y_{p+1} = \sum(-a_i/a) y_i \in P(T)$, a contradiction; so $a = 0$ and all $a_i = 0$. ∎

Theorem 8.11 (50.5, imbedding theorem)

Every compact metrizable space X of topological dimension m imbeds in $\mathbb{R}^{2m+1}$.

Proof. Let $N = 2m+1$, $|\cdot|$ the square metric on $\mathbb{R}^N$, and $\mathcal{C}(X, \mathbb{R}^N)$ with the sup metric ρ, complete and hence Baire. For continuous f let $\Delta(f) = \sup\{\operatorname{diam} f^{-1}(z) : z \in f(X)\}$ and $U_\epsilon = \{f : \Delta(f) < \epsilon\}$.

U_ϵ is open: let $\Delta(f) < b < \epsilon$. The set $A = \{(x, y) : d(x, y) \geq b\}$ is compact and $|f(x) - f(y)| > 0$ on it, with positive minimum δ (if $A \neq \emptyset$). If $\rho(f, g) < \delta/2$, then $g(x) \neq g(y)$ on A, so $\Delta(g) \leq b < \epsilon$.

U_ϵ is dense: let f and $\delta > 0$ be given. Cover X by open sets of diameter less than $\epsilon/2$ whose images under f have diameter less than $\delta/2$; by compactness and $\dim X = m$ there is a finite open refinement $\{U_1, \dots, U_n\}$ covering X of order at most $m+1$, and a partition of unity $\{\phi_i\}$ dominated by it (Theorem 4.16). Choose $x_i \in U_i$ and, by Lemma 8.10, points z_i in general position with $|z_i - f(x_i)| < \delta/2$. Let $g = \sum \phi_i z_i$. If $\phi_i(x) > 0$, then $x \in U_i$ and $|f(x) - z_i| < \delta$; since $f(x) - g(x) = \sum \phi_i(x)(f(x) - z_i)$, $|f(x) - g(x)| < \delta$, and by compactness $\rho(f, g) < \delta$. If $g(x) = g(y)$, then $\sum(\phi_i(x) - \phi_i(y)) z_i = 0$ with coefficients summing to 0, and only indices with $x \in U_i$ or $y \in U_i$ appear, at most $2(m+1) = N+1$ of them. By general position all coefficients vanish. Choosing i with $\phi_i(x) > 0$ gives $\phi_i(y) > 0$, so $x, y \in U_i$ and $d(x, y) < \epsilon/2$. Hence $\Delta(g) \leq \epsilon/2$ and $g \in U_\epsilon$.

By Lemma 8.1, $\bigcap_n U_{1/n}$ is dense, hence nonempty; a map g in it is injective, hence an imbedding (Theorem 3.20). ∎

Theorem 8.12 (50.6)

Every compact subspace of $\mathbb{R}^N$ has topological dimension at most N.

Proof.

Step 1. an open covering of $\mathbb{R}^N$ by sets of square-metric diameter at most 1, of order at most $N+1$. The cubical cells of $\mathbb{R}^N$ are the products $\sigma = \prod I_i$ with each I_i either $\{k_i\}$ ($k_i \in \mathbb{Z}$) or $(k_i, k_i + 1)$; let $P(\sigma)$ be the set of indices of the first kind. Every point lies in exactly one cell. Write $t_i(y)$ for the distance from y_i to $\mathbb{Z}$, and let

$$V(\sigma) = \Big\{y : \ |y_i - k_i| < \tfrac{1}{2} \ (i \in P(\sigma)), \quad y_i \in (k_i, k_i + 1) \ (i \notin P(\sigma)), \quad \max_{i \in P(\sigma)} t_i(y) < \min_{i \notin P(\sigma)} t_i(y)\Big\},$$

an open set of diameter at most 1 containing σ. If $\sigma \neq \tau$ and $|P(\sigma)| = |P(\tau)|$, then $V(\sigma) \cap V(\tau) = \emptyset$: if $P(\sigma) = P(\tau)$ some k_i differs, which is impossible on a point coordinate (nearest integer) and on an interval coordinate (disjoint intervals); otherwise there are $i \in P(\sigma) - P(\tau)$ and $j \in P(\tau) - P(\sigma)$, and a common point would satisfy $t_i < t_j$ and $t_j < t_i$. So each point lies in at most one $V(\sigma)$ for each value $|P(\sigma)| \in \{0, \dots, N\}$, hence in at most $N+1$ of them.

Step 2. let $X \subset \mathbb{R}^N$ be compact, and let $\mathcal{A}$ be an open covering of X with Lebesgue number δ (Lemma 3.28). Choose $0 < s < \delta$; the sets $sV(\sigma) \cap X$ have diameter less than δ, so each lies in an element of $\mathcal{A}$; they form an open refinement covering X of order at most $N+1$. ■

Corollary 8.13 (50.7)

Every compact m-manifold has topological dimension at most m.

Proof. Each x has an open U with a homeomorphism $h : U \to W \subset \mathbb{R}^m$ open. Choose a closed ball $\overline{B} \subset W$ about $h(x)$; then $C_x = h^{-1}(\overline{B})$ is compact (so closed in X), contains the neighbourhood $h^{-1}(B)$ of x, and has dimension at most m by Theorem 8.12. Finitely many C_x cover X; apply Corollary 8.9. ■

Corollary 8.14 (50.8)

Every compact m-manifold imbeds in $\mathbb{R}^{2m+1}$.

Proof. It is compact Hausdorff, hence regular, with a countable basis, so metrizable (Theorem 4.12); its dimension k is at most m, and it imbeds in $\mathbb{R}^{2k+1} \subset \mathbb{R}^{2m+1}$ by Theorem 8.11. ■

Corollary 8.15 (50.9)

A compact metrizable space imbeds in some $\mathbb{R}^N$ iff it has finite topological dimension.

Proof. ($\Rightarrow$) Theorem 8.12.

($\Leftarrow$) Theorem 8.11. ■

Without compactness: every m-manifold, and more generally every metrizable space of dimension m with a countable basis, imbeds in $\mathbb{R}^{2m+1}$ (Menger–Nöbeling).

8.4 Selected exercises

Motivation. Exercises from Munkres on Baire's theorem in locally compact spaces, continuity points of functions on $\mathbb{R}$, and dimension and connectedness.

Exercise 8.1 (§48, Ex. 3; locally compact spaces are Baire)

Show that every locally compact Hausdorff space X is a Baire space.

Solution. Let $A_1, A_2, \ldots$ be closed with empty interior and U_0 nonempty open. We show $U_0 \not\subset \bigcup_n A_n$.

Step 1. Since A_1 has empty interior, $U_0 - A_1$ is a nonempty open set. Pick a point in it; by Theorem 3.35 it has a neighbourhood U_1 with $\overline{U_1}$ compact and $\overline{U_1} \subset U_0 - A_1$. Inductively, given a nonempty open U_{n-1}, choose a nonempty open U_n with $\overline{U_n}$ compact and $\overline{U_n} \subset U_{n-1} - A_n$.

Step 2. The sets $\overline{U_n}$ are nonempty, closed in the compact space $\overline{U_1}$, and nested, so they have the finite intersection property and $\bigcap_n \overline{U_n} \neq \emptyset$ (Theorem 3.23).

Step 3. A point x of the intersection lies in $\overline{U_1} \subset U_0$, and for each n, $x \in \overline{U_n} \subset U_{n-1} - A_n$, so $x \notin A_n$. Hence $U_0 \not\subset \bigcup A_n$: the union has empty interior. ■

Exercise 8.2 (§48, Ex. 7; continuity exactly on a countable dense set)

Let D be a countable dense subset of $\mathbb{R}$. Show that there is no $f : \mathbb{R} \to \mathbb{R}$ that is continuous precisely at the points of D.
(a) Show that the set C of points at which a function $f : \mathbb{R} \to \mathbb{R}$ is continuous is a G_δ set.
(b) Show that D is not a G_δ set.

Solution.

(a) For $U \subset \mathbb{R}$ let $\operatorname{diam} f(U) = \sup\{|f(s) - f(t)| : s, t \in U\}$, and let U_n be the union of all open U with $\operatorname{diam} f(U) < 1/n$; U_n is open.

Step 1. $C \subset \bigcap U_n$: if f is continuous at x, choose $\delta > 0$ with $|f(y) - f(x)| < 1/(3n)$ for $|y - x| < \delta$. Then $U = (x - \delta, x + \delta)$ has $\operatorname{diam} f(U) \le 2/(3n) < 1/n$, so $x \in U_n$.

Step 2. $\bigcap U_n \subset C$: let $x \in \bigcap U_n$ and $\epsilon > 0$; choose n with $1/n < \epsilon$. Then x lies in an open U with $\operatorname{diam} f(U) < 1/n$, so $|f(y) - f(x)| < \epsilon$ for all $y \in U$. Thus f is continuous at x, and $C = \bigcap_n U_n$.

(b) Suppose $D = \bigcap_n W_n$ with W_n open. Each $W_n \supset D$ is dense, and for $d \in D$ the open set $V_d = \mathbb{R} - \{d\}$ is dense. The family $\{W_n\} \cup \{V_d : d \in D\}$ is countable, and

$$\bigcap_n W_n \cap \bigcap_{d \in D} V_d = D - D = \emptyset.$$

But $\mathbb{R}$ is a complete metric space, hence a Baire space (Theorem 8.2), so this countable intersection of dense open sets is dense by Lemma 8.1, a contradiction.

By (a) and (b), the set of continuity points of f is never D. ■

Exercise 8.3 (§50, Ex. 2; connected spaces have dimension at least 1)

Show that a connected T_1 space X with more than one point has dimension at least 1.

Solution. If X is not finite dimensional there is nothing to prove, so suppose $\dim X = 0$ and derive a contradiction.

Step 1. Choose $x \neq y$. As X is T_1, $\mathcal{A} = \{X - \{x\}, X - \{y\}\}$ is an open covering. Since $\dim X = 0$, it has an open refinement $\mathcal{B}$ covering X of order at most 1, i.e. the elements of $\mathcal{B}$ are pairwise disjoint.

Step 2. Let $V \in \mathcal{B}$ contain x. V lies in an element of $\mathcal{A}$, which must be $X - \{y\}$ since $x \in V$; so $y \notin V$. Let W be the union of the other elements of $\mathcal{B}$.

Step 3. V and W are open, disjoint, cover X, and are nonempty ($x \in V$, $y \in W$). This is a separation of the connected space X, a contradiction. ■